

WSQ COURSE • TGS-2023036153

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Multi AI Agents Workflow for Content Creation

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Digital Attendance



Attendance is mandatory for SSG funding.

Take AM, PM and Assessment digital attendance for every session.

Use the SSG / TMS QR code provided by your trainer.

**Minimum
75%
attendance
required**

About the Trainer



[Trainer Name]

[Title / Role]



Qualifications: _____



Expertise: _____



Experience: _____



Contact / Profile: _____

About the Trainer



Dr. Alfred Ang

AI Trainer & Engineer



PhD-qualified trainer specialising in AI, software engineering and developer tooling.



Builds multi-agent content workflows with Claude, Codex and Hermes Agent.



Trains professionals across Singapore on practical, project-based AI skills.



Certifications: PMP, AWS, Microsoft, CompTIA, CKA, CKAD.



[linkedin.com/in/angchewhoe](https://www.linkedin.com/in/angchewhoe)



github.com/alfredang

INTRODUCTION

Let's Know Each Other

Take a minute to introduce yourself to the class:



Your name & role



What you do day to day



Your content work today



What you want from this
course

Ground Rules



Phones on silent

Step out quietly for calls or breaks.



Participate actively

No question is stupid — ask freely.



Mutual respect

Agree to disagree; one conversation at a time.



Be punctual

Return from breaks on time.



75% attendance

Required for funding eligibility.



Stay hands-on

Follow along with every activity.

Learning Outcomes

By the end of the two days you will be able to:

L01 · Conceptualise

Conceptualise content ideas that meet marketing objectives, and map them into digital storyboards.

L02 · Identify

Identify content requirements from customer preferences, and decide how often to publish.

L03 · Determine

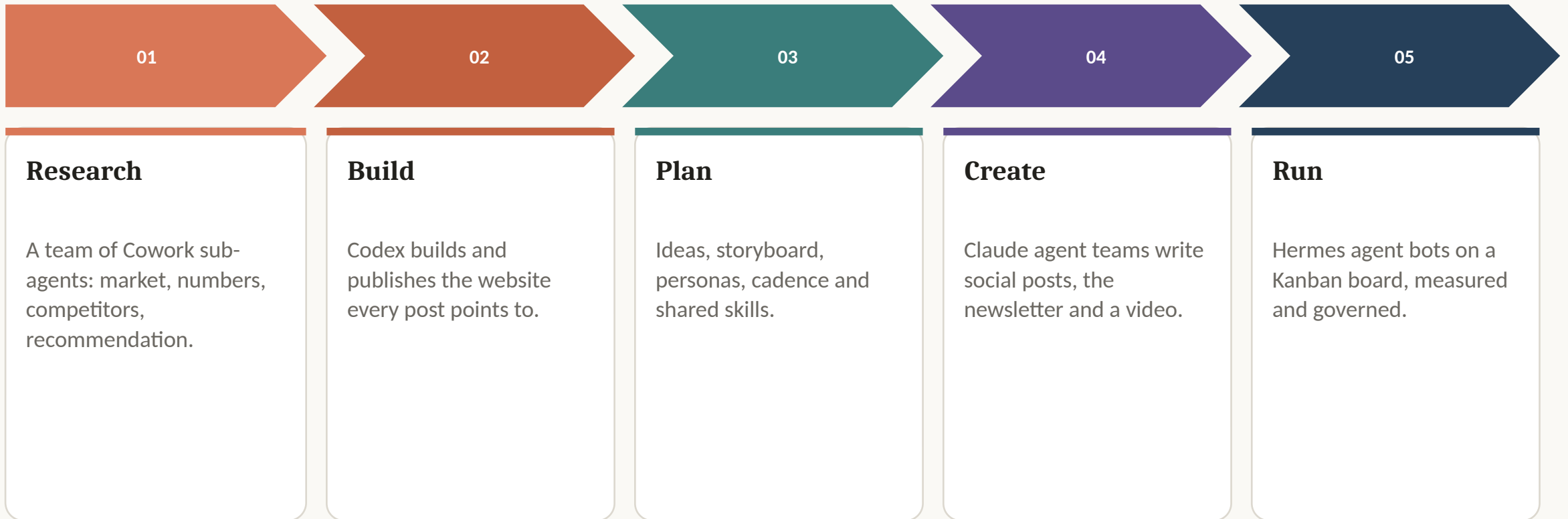
Determine content types and styles, and the modes and processes for distributing content.

L04 · Develop

Develop guidelines for executing the content strategy, with suitable delivery modes and responsible AI practices.

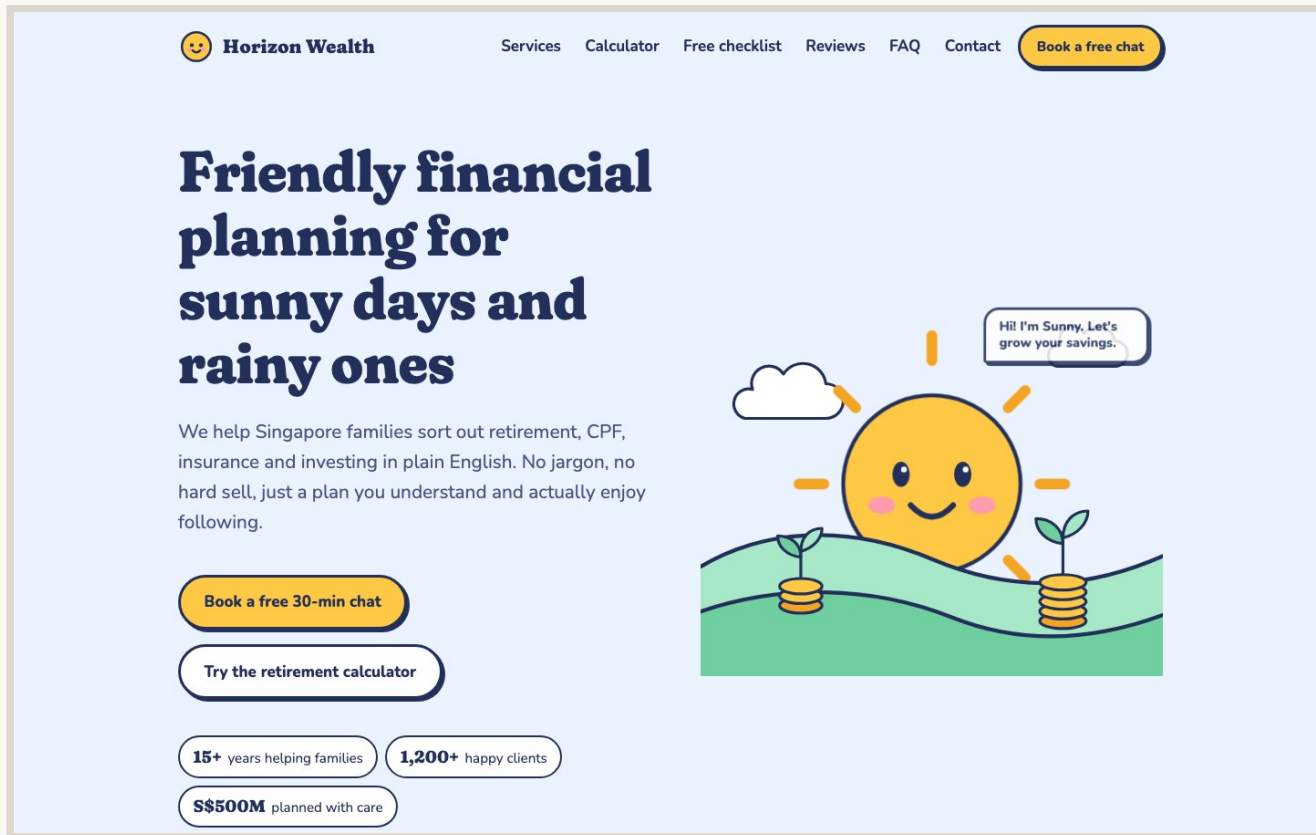
One Firm, One Content Studio of Agents

Horizon Wealth Planning — independent financial planners in Singapore since 2010. Goal: 40 booked free chats a month by March 2027.



Meet Horizon Wealth Planning

The firm's website — the reference build for Lab 2. Every post, email and video in this course points back here.



Plain English

Retirement, CPF, insurance and investing — no jargon, no hard sell.

One goal

Turn visitors into booked free 30-minute chats.

Regulated

Financial content: general information only, never a promised return.

The Right Tool for Each Job

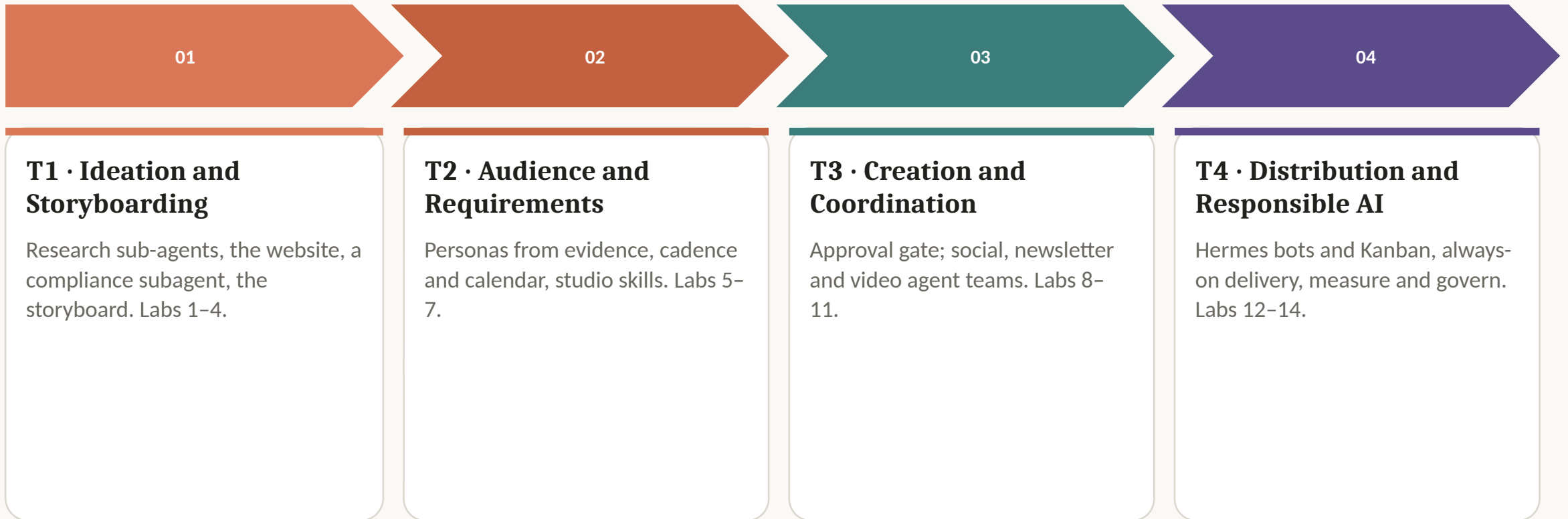
Four tools, one studio. Pick by the job, not by habit.

TOOL	USE IT FOR	IN THIS COURSE
Claude Cowork	Business work on your files and apps: research, analysis, skills, email drafts	Labs 1, 4–7, 10, 14
Codex	Building software: the website, scripts, the approval gate, the video	Labs 2–3, 8, 11
Claude Code	Subagent definitions and agent teams that write and review together	Labs 9–11
Hermes Agent	Open-source agent bots on any model, a Kanban board, schedules and chat apps	Labs 12–13

TIP Every external post, email and video passes one gate: a person approves the exact version.

Course Outline

Four topics over two days. Every topic ends in labs that move Horizon forward.



Lab Materials

Fourteen labs, each in its own folder with a README, prompts (MD and PDF), assets and an evidence checklist.

DAY	LABS	TOOL
1	1 Research team • 2 Build the site • 3 Compliance + publish	Cowork, Codex
1	4 Ideas + storyboard • 5 Personas • 6 Cadence • 7 Skills	Cowork
2	8 Approval gate • 9 Social team • 10 Newsletter team	Codex, Claude Code, Cowork
2	11 Video team • 12 Hermes Kanban • 13 Telegram bot	Claude Code, Codex, Hermes
2	14 Measure, govern, playbook	Cowork

TIP Labs build on each other: one horizon-studio folder from Lab 1 to Lab 14.

SCHEDULE

Lesson Plan (9:00 AM – 6:00 PM)

Tea breaks 10 min, lunch 45 min. Full timings and slide numbers are in the Lesson Plan.

BLOCK	DAY 1 — RESEARCH, BUILD, PLAN	DAY 2 — CREATE, DISTRIBUTE, GOVERN
Morning 1	9:00–10:45 Welcome • Topic 1 • Lab 1	9:00–10:45 Topic 3 • Labs 8–9
Morning 2	10:55–12:40 Labs 2–3	10:55–12:30 Labs 10–11 • recap
Afternoon 1	13:25–15:25 Lab 4 • Topic 2 • Lab 5	13:15–14:50 Topic 4 • Labs 12–13
Afternoon 2	15:35–18:00 Labs 6–7 • review	15:00–16:00 Lab 14 • summary • 16:00–18:00 Assessment

TIP Day 2 assessment: Written Assessment 4:00–5:00 PM, Practical Performance 5:00–6:00 PM.

Briefing for Assessment



Clear your space

Place phones and other materials under the table or on the floor.



No recording

No photos or recording of assessment materials.



Open book

Only slides, learner guide and approved materials are allowed.



Work individually

Complete the assessment on your own; no discussion.



Manage your time

Written Assessment then Practical Performance — watch the clock.



Appeals

An appeal process is available if you disagree with the outcome.

Assessment & Funding



Final Assessment

- Written Assessment (WA) — 1 hour, starts 4:00 PM
- Practical Performance (PP) — 1 hour
- Format: Open book — class ends 6:00 PM
- Allowed: slides, learner guide & approved materials only
- Appeal process available



Criteria for Funding

Minimum 75% attendance

Based on SSG digital attendance records.

Assessed as 'Competent'

Complete and pass both assessment components.

Courseware & Assessment on the LMS



Download the courseware

Get the slides and learner guide from the LMS for the open-book assessment.



Take the assessment

Complete the Written Assessment and Practical Performance on the LMS.



Log in

Use your enrolled account to access the course materials and assessment.

A screenshot of the Tertiary Infotech Academy LMS login interface. It features a dark blue background with a white logo at the top. Below the logo, the text "Tertiary Infotech Academy" and "LMS cum TMS for WSQ Courses" is displayed. There is an "Email" input field, a "Send OTP" button, and a link "Login with password instead". At the bottom, there are links for "Privacy Policy", "Acceptable Use Policy", and "Feedback", along with the text "Powered by Tertiary Infotech Academy Pte Ltd".


Tertiary Infotech Academy
LMS cum TMS for WSQ Courses

Email

Send OTP

[Login with password instead](#)

[Privacy Policy](#) [Acceptable Use Policy](#) [Feedback](#)
Powered by Tertiary Infotech Academy Pte Ltd

LMS: <https://lms-tms.tertiaryinfotech.com/>

Certification & TRAQOM Survey



TRAQOM Survey (Mandatory)

Complete the certification & TRAQOM survey to receive your certificate.

<https://lms-tms.tertiaryinfotech.com/>



Digital Attendance

Remember to take AM, PM and Assessment digital attendance — required for your funding.

Multi-AI-Agent Content Ideation and Digital Storyboarding

- 1 From one chatbot to teams of agents
- 2 Claude Cowork, Codex, Claude Code and Hermes — and the agent loop
- 3 Research with sub-agents; build and check the website
- 4 Ideas from parallel agents, then a five-beat storyboard

From Prompts to Agent Teams

Prompt Engineering

2023

ChatGPT goes mainstream. Craft the wording of one prompt.

Tools and MCP

2024

Models call tools. Anthropic open-sources the Model Context Protocol (Nov 2024).

Context Engineering

2024

Fill the context with exactly what the next step needs: files, memory, retrieval, skills.

Harness Engineering

2025

Agents ship inside a harness — Claude Code, Codex. The loop, tools and checks around the model.

Agent Teams

2026

Many agents with their own roles: Claude Code agent teams, Codex subagents, Hermes Agent and its Kanban.

Years show when each practice took hold; the names came later. One agent is a helper. A team with a reviewer and a human gate is a studio.

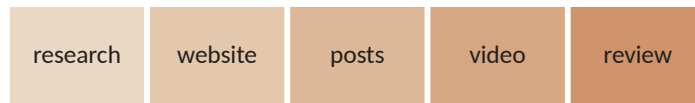
Why a Team of Agents?

Ask for sub-agents and the main agent runs specialists in parallel, each in its own context, then combines what they return.

ONE AGENT DOES EVERYTHING

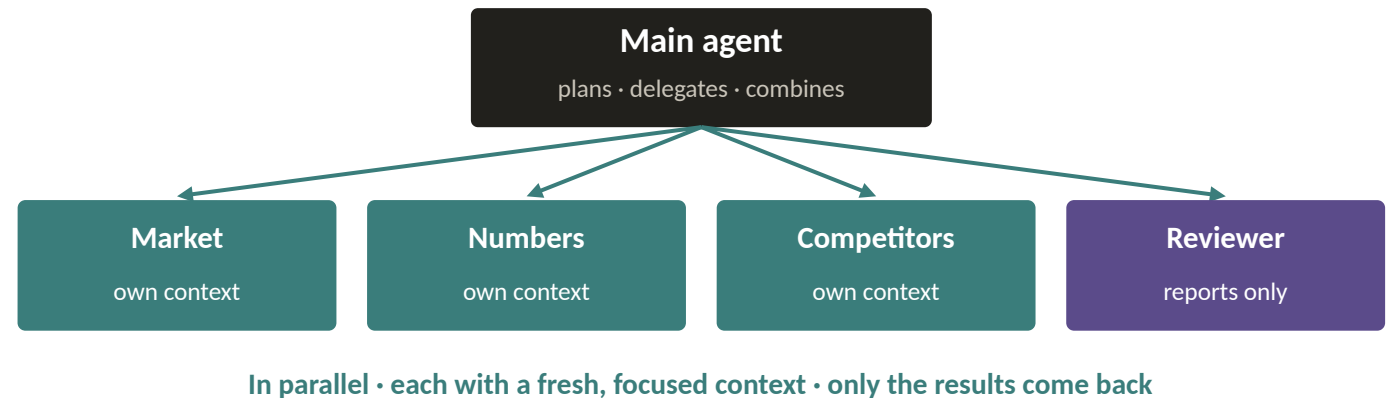
One agent

One context window,
filling up with every job



One job after another • one voice • marks its own homework

A MAIN AGENT DELEGATES



Faster

Independent research runs at the same time, not one after another.

Focused

Each agent holds only its own job — no context full of other people's work.

Specialised

Its own role, tools, model and rules.

Second opinion

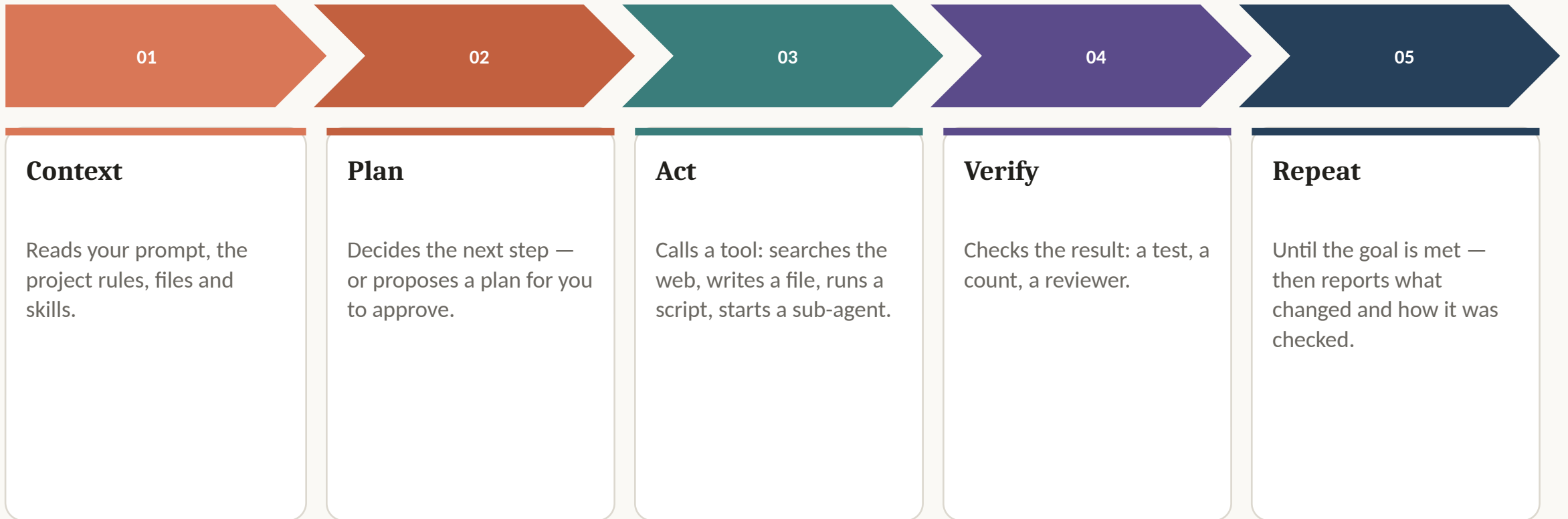
A reviewer that did not write the post checks it.

The cost

More tokens and more to review. Split only work that is truly independent.

The Agent Loop

Every tool in this course runs the same loop. Sub-agents and teams are more loops, running side by side.



Four Tools, One Studio

All four run agents. They differ in what they are built for.

Claude Cowork

In the Claude desktop app. Works on a folder and your connected apps; builds Live Artefacts; runs sub-agents in parallel.

Codex

In the ChatGPT desktop app. Builds software in a project: /plan, AGENTS.md, subagents, hooks.

Claude Code

Terminal, IDE or desktop. Subagent files in .claude/agents/ and agent teams with a shared task list.

Hermes Agent

Open source (MIT) from Nous Research. Any model; profiles as bots; a Kanban board; cron; Telegram.

Get the Tools Ready

Install once; the labs tell you which tool to open.

DOWNLOAD

claude.com/download

Claude desktop

Cowork (and Claude Code) — Pro plan or above

ChatGPT desktop

Codex — chatgpt.com/download

Claude Code needs a Pro, Max, Team, Enterprise or Console account.

1

Claude desktop Install from claude.com/download and sign in on a paid plan. Cowork is in the sidebar.

2

Claude Code CLI `curl -fsSL https://claude.ai/install.sh | bash` (Windows: `irm https://claude.ai/install.ps1 | iex`).

3

ChatGPT desktop For Codex. Sign in on a plan that includes Codex.

4

Hermes Agent Lab 12 — installed in class with the trainer's API key.

5

Node.js 22, Python 3, Git For the scripts, the video and GitHub Pages.

Claude Cowork and Sub-agents

Cowork brings Claude's agent loop to everyday business work.

Point it at a folder

Cowork reads and writes the files in horizon-studio, and asks before it changes anything important.

Ask for sub-agents

Say "use sub-agents, one per role, in parallel". Each works in its own context; only results come back.

Get a file, not a chat

Reports, CSVs, Live Artefacts and email drafts — saved where the next lab can use them.

The Research Team

Lab 1: four specialists, three in parallel, one to decide what it means.



Research You Can Trust

Agents write plausible numbers fluently. Make them prove each one.

Cite or say UNKNOWN

Every figure needs a source and a date. A gap marked UNKNOWN beats a confident guess.

Keep internal data internal

services.csv has INTERNAL fee columns for analysis. They never appear in anything public.

Read the sources

Open two of the links yourself. If a source does not say it, the report does not either.

A Research Team of Sub-agents

LAB 01

DAY 1 • ~45 MIN

THE STORY SO FAR

October 2026. Rachel wants 40 booked chats a month by March. Before she spends a dollar, her partners want three answers: who to reach, what the funnel needs, and what competitors already publish.

YOU'LL BUILD

research/: market research, business analysis, competitive analysis and a recommendation, every figure cited

Surface: Claude Cowork

Features: sub-agents in parallel • market research • business analysis • competitive analysis • a recommendation

A Research Team of Sub-agents

Claude Cowork • full step-by-step with reasoning is in your Learner Guide.

- 1 Make the studio folder** Create horizon-studio with a data/ folder. Copy this lab's assets into data/.
- 2 Open Cowork** In the Claude desktop app choose Cowork, then point it at the horizon-studio folder.
- 3 Brief the team** Paste Prompt A. Cowork confirms the four roles and the plan — correct anything before it starts.
- 4 Run them in parallel** Paste Prompt B. Watch three sub-agents work at once in the progress panel.
- 5 Audit the sources** Open competitive-analysis.md. Every figure needs a source and a date; gaps say UNKNOWN.
- 6 Decide** Read recommendation.md. Change one thing you disagree with — it is your recommendation now.

A Research Team of Sub-agents

PROMPT A — Cowork: brief the team

I'm the marketing lead at Horizon Wealth Planning, an independent financial-planning firm in Singapore. Read data/business-brief.md and data/research-agents-spec.md.

We will run a research team of four sub-agents:

1. market-researcher — Singapore demand for financial planning: who, which channels, why now.
2. business-analyst — our funnel from data/firm-metrics.csv: what 40 chats a month needs.
3. competitive-analyst — six named competitors:
offer, pricing model, channels, content themes.
4. strategy-advisor — reads the other three and writes the recommendation.

Rules for all: cite every figure with source and date, or write UNKNOWN. Never publish the INTERNAL columns. Confirm each role in one line and show me your plan. Do not start yet.

A Research Team of Sub-agents (cont.)

PROMPT B — Cowork: run the team

Run the research team now, using sub-agents.

- Start market-researcher, business-analyst and competitive-analyst in parallel, each as its own sub-agent, each saving its report to research/.
- When all three are done, the strategy-advisor writes research/recommendation.md exactly as research-agents-spec.md describes. Then give me a five-line summary and a list of everything marked UNKNOWN.

NOTE A sub-agent does its research in its own context and hands back only the result. Your main conversation stays clear for the decision.

A Research Team of Sub-agents

Check your work before moving on.

CHECK YOUR WORK

- ✓ Cowork confirmed all four roles before starting.
- ✓ Three research sub-agents ran at the same time.
- ✓ research/ holds market-research, business-analysis, competitive-analysis and recommendation.
- ✓ The competitive analysis names six real competitors, each with a source and a date.
- ✓ The business analysis shows conversion rates and the visits needed for 40 chats a month.
- ✓ No uncited figure; no INTERNAL fee anywhere in the reports.

Codex: Plan Before You Build

Create a local project, add the horizon-studio folder, then plan.

/plan first

Codex reads the brief and asks what it cannot determine. Every question is a guess it was about to make.

Build from the brief

site-brief.md, brand.md and the CSVs — not the model's idea of a financial website.

AGENTS.md

/init drafts the project rules. Trim to What this is, Commands, Conventions, Boundaries.

The Retirement Calculator

The acceptance test: age 30, retire 62, S\$20,000, S\$800 a month, 4% → S\$693,138.

The screenshot shows the Horizon Wealth Retirement Calculator interface. The header includes the Horizon Wealth logo, navigation links (Services, Calculator, Free checklist, Reviews, FAQ, Contact), and a 'Book a free chat' button. The main heading is 'How big could your nest egg grow?' with a subtext: 'Play with the numbers to see a rough picture of your savings at retirement. It takes 30 seconds, and nothing you type leaves your browser.'

The calculator form includes the following fields and controls:

- Your age now:** Input field with value 30.
- Age you'd like to retire:** Input field with value 62.
- Savings so far (not CPF):** Input field with value S\$ 20000.
- Saved each month:** Input field with value S\$ 800.
- Average yearly return (not guaranteed):** A slider control set to 4%.

The results section displays:

- At 62 you could have about:** **S\$693,138**
- That's roughly S\$2,310 a month over a 25-year retirement, on top of your CPF LIFE payouts.**
- A bar chart showing the growth of savings at ages 36, 43, 49, 56, and 62.
- A button: **Ask a planner to check my numbers**
- A disclaimer: **Illustration only. Assumes a steady return and fixed monthly savings, before inflation and fees. Returns are not guaranteed and this is not financial advice.**

A small yellow circular button with an upward arrow is located at the bottom right of the calculator form.

Monthly compounding

Rate ÷ 12, years × 12. The test number proves the maths.

Labelled

"Illustration only ... not guaranteed ... not financial advice." Word for word.

Private

Nothing typed leaves the browser.

Checklist and Contact Form

The two places a visitor becomes a lead.

Free: the 10-minute Money Check-up Checklist

Ten quick questions every Singapore household should be able to answer. Tick them off and you'll know exactly where you stand, and what to fix first.

- ☒ Is your emergency fund big enough?
- ☒ Are your CPF nominations and will up to date?
- ☒ Do you have gaps (or overlaps) in your insurance?
- ☒ Are you using SRS to cut your income tax?
- ☒ Do you know your retirement number?

Instant access, plus one friendly money tip a month. No spam, unsubscribe any time.

Get your free checklist

Pop in your email and the checklist unlocks right here, ready to tick off or print.

Unlock my free checklist

A lead magnet

The 10-item Money Check-up, unlocked with an email.

Consent

The contact form will not submit without the PDPA consent box.

No backend

A thank-you message in class; a form service in production.

Plan and Build the Horizon Website with Codex

LAB 02

DAY 1 • ~45 MIN

THE STORY SO FAR

The research is back: mid-career families and pre-retirees, on LinkedIn, Facebook and email. Every post will point to the website — and the current one is out of date. Rachel wants a new site this week, built from the brief so nothing is invented.

YOU'LL BUILD

docs/index.html: hero, services, retirement calculator, checklist, reviews, FAQ, contact form with PDPA consent

Surface: Codex

Features: Add a folder • /plan • AGENTS.md (/init) • build from data, not memory

Plan and Build the Horizon Website with Codex

Codex • full step-by-step with reasoning is in your Learner Guide.

- 1 Add the brief** Copy the assets into horizon-studio/data/. No Lab 1 recommendation? Copy recommendation-sample.md to research/recommendation.md.
- 2 Open it in Codex** In the ChatGPT desktop app open Codex, create a local project and add the horizon-studio folder. Run git init if it is not a repo yet.
- 3 Plan first** Type /plan, then paste the prompt. Read every question Codex asks.
- 4 Answer and narrow** Answer the questions. Cut one step you did not ask for, then approve.
- 5 Serve it** Run `python3 -m http.server 8080 --directory docs` and open `http://localhost:8080`.
- 6 Test it** Calculator: 30, 62, S\$20,000, S\$800, 4% must show S\$693,138. Try it at 375px wide.
- 7 Write the rules** Run /init. Trim AGENTS.md to What this is, Commands, Conventions, Boundaries — under 60 lines.

Plan and Build the Horizon Website with Codex

PROMPT — Codex, after /plan

Build the Horizon Wealth Planning website in docs/index.html (GitHub Pages will serve docs/). Read data/site-brief.md, data/brand.md, data/services.csv, data/faq.md, data/testimonials.csv, data/checklist-items.md and research/recommendation.md first.

MUST HAVE

- Every section in site-brief.md, in that order.
- Six service cards — never the INTERNAL columns.
- A retirement calculator with monthly compounding and the illustration note word for word.
- The checklist unlocked by email; contact form with a REQUIRED PDPA consent box; no backend.
- The footer disclaimer word for word.

CONSTRAINTS

- One HTML file, inline CSS and JS, no framework.
- Accessible: labels, alt text, focus, contrast.

DONE WHEN

- The calculator test gives S\$693,138.
- No horizontal scroll at 375px.

NOTE Build from the brief, not from memory. The model knows what a financial website looks like; only the brief knows what Horizon is allowed to say.

Plan and Build the Horizon Website with Codex

Check your work before moving on.

CHECK YOUR WORK

- ✓ Plan mode showed steps AND questions before any edit.
- ✓ You answered its questions instead of letting it guess.
- ✓ The calculator test shows S\$693,138 with its illustration note.
- ✓ Six service cards; no INTERNAL fee or hours anywhere.
- ✓ The contact form will not submit without consent.
- ✓ No horizontal scroll at 375px; AGENTS.md is under 60 lines.

Financial Content Is Regulated

Twelve rules every Horizon page, post, email and video must pass (compliance-checklist.md).

RULE	WHAT IT MEANS	IF BROKEN
C1 • Not advice	General information only — say so	Critical
C2 • No promises	No guaranteed or "risk-free" returns	Critical
C3 • Label illustrations	Assumptions shown; not guaranteed	Critical
C5 • No performance	Never client returns or "average" returns	Critical
C9 • Consent	Marketing only to people who opted in	Critical
C11 • Facts sheet	Every figure from facts-2026.md	Critical

TIP MAS Guidelines on Standards of Conduct for Digital Advertising Activities apply from 25 March 2026.

A Reviewer That Only Reports

Lab 3: Codex starts a compliance reviewer as a subagent.

The writer (main agent)

- Built the page and wants it to pass
- Can edit every file
- Fixes what the reviewer reports
- Re-runs the reviewer until PASS

The reviewer (subagent)

- Did not write the page
- Reads only — never edits
- Table: words, rule, severity, fix
- PASS only with no Critical or High issue

A Compliance Subagent, Then Publish

LAB 03

DAY 1 • ~35 MIN

THE STORY SO FAR

The site looks great. Then Rachel's partner reads the 'How it works' intro she drafted — '8% a year ... guaranteed' — and goes pale. Before the site goes live, a reviewer that did not write it must check it.

YOU'LL BUILD

review/site-compliance.md (FAIL, then PASS) and the site live on GitHub Pages

Surface: Codex → GitHub Pages

Features: a reviewer subagent that only reports • the compliance checklist • GitHub Pages from /docs

A Compliance Subagent, Then Publish

Codex → GitHub Pages • full step-by-step with reasoning is in your Learner Guide.

- 1 Add the rules** Copy compliance-checklist.md and facts-2026.md into data/.
- 2 Run the reviewer** Paste Prompt A. Codex starts the reviewer as a subagent — watch it in the Subagents panel.
- 3 Read the report** It must flag Rachel's "8% a year ... guaranteed" line as Critical. Note anything else it found.
- 4 Fix and re-check** Paste Prompt B. The main agent fixes; the reviewer checks again until PASS.
- 5 Publish** Create the horizon-studio repo on GitHub, then Settings → Pages → Deploy from a branch → main, /docs.
- 6 Test on a phone** Open the Pages URL on your phone and try the calculator and the contact form.

A Compliance Subagent, Then Publish

PROMPT A — Codex: the reviewer

Use a subagent as our compliance reviewer. Its job: check docs/index.html against data/compliance-checklist.md and data/facts-2026.md. It reports; it never edits a file.

Output a table — where, the exact words, the rule (C1-C12), severity (Critical / High / Low), the fix — and end with PASS or FAIL. FAIL if any Critical or High issue remains. Save the report to review/site-compliance.md.

A Compliance Subagent, Then Publish (cont.)

PROMPT B — Codex: fix, re-check, ship

Fix every Critical and High issue in review/site-compliance.md, then run the compliance reviewer subagent again. Repeat until it says PASS.

Then commit and push to GitHub. Before committing, list every file you will add and confirm there is no .env, key or token among them. Give me the Pages URL: <https://<your-user>.github.io/horizon-studio/>

NOTE The writer should never mark its own homework. A reviewer that cannot edit has no reason to go easy on the draft.

A Compliance Subagent, Then Publish

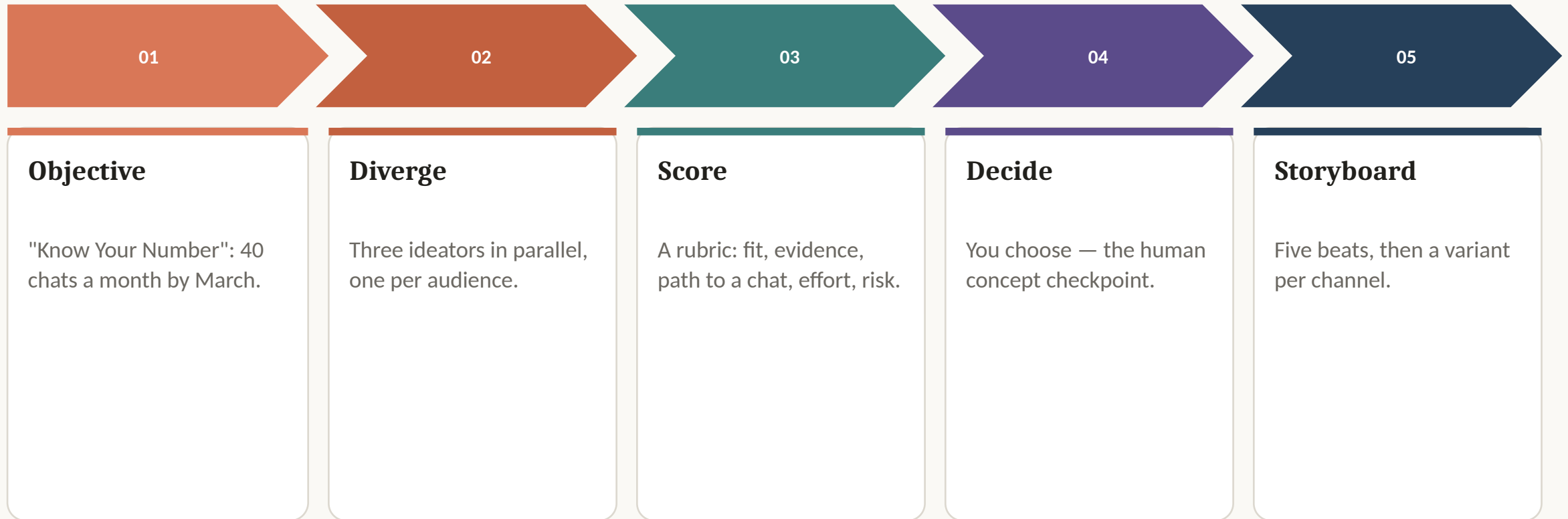
Check your work before moving on.

CHECK YOUR WORK

- ✓ The reviewer ran as a subagent and edited nothing.
- ✓ The first report flagged the "8% ... guaranteed" line as Critical (C2, C5).
- ✓ The final report says PASS.
- ✓ The footer disclaimer and calculator note are word for word.
- ✓ Codex listed the files before committing; no secret among them.
- ✓ The Pages URL works on your phone.

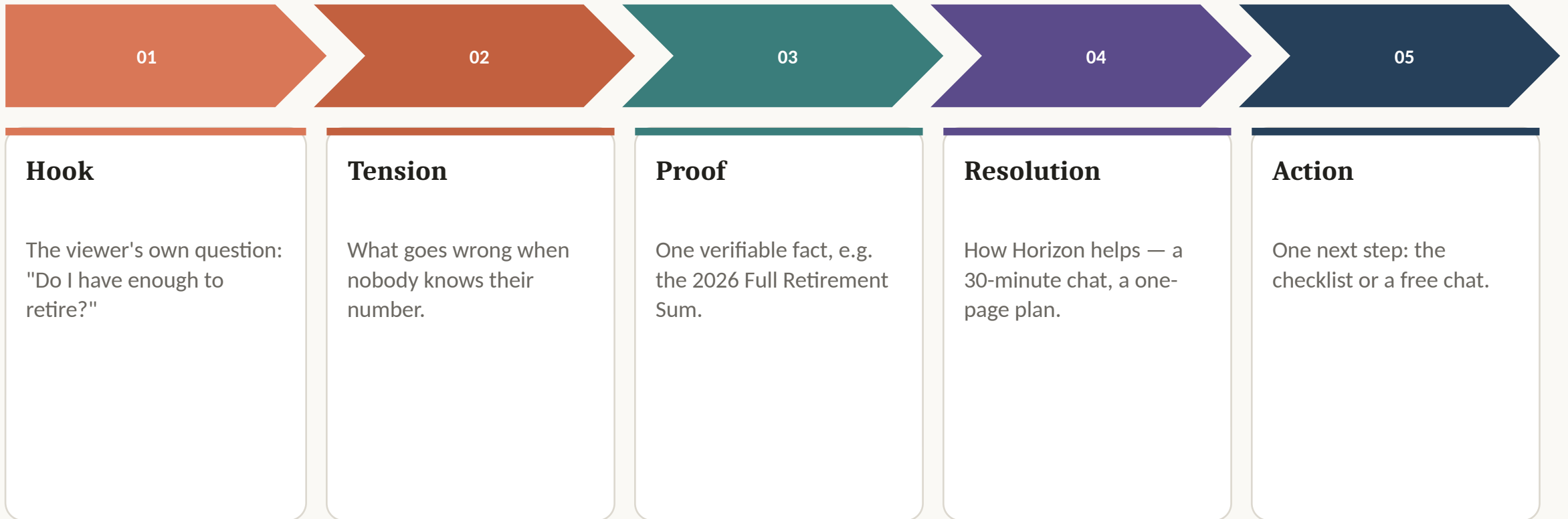
From Objective to Storyboard

Agents diverge; a person decides; agents make it concrete.



The Five-Beat Storyboard

Every channel tells the same story with the same claims.



Ideation Sub-agents and a Digital Storyboard

LAB 04

DAY 1 • ~40 MIN

THE STORY SO FAR

The site is live. Now the campaign: 'Know Your Number', November to January. Rachel wants ideas for three very different audiences — and a storyboard so LinkedIn, Facebook and YouTube tell the same story.

YOU'LL BUILD

strategy/ideas.md (15 scored ideas), strategy/storyboard.md and an interactive storyboard Live Artefact

Surface: Claude Cowork (sub-agents, Live Artefact)

Features: three ideators in parallel • rubric scoring • a human concept checkpoint • a five-beat storyboard

Ideation Sub-agents and a Digital Storyboard

Claude Cowork (sub-agents, Live Artefact) • full step-by-step with reasoning is in your Learner Guide.

- 1 Add the brief** Copy the assets into data/ and open Cowork on horizon-studio.
- 2 Diverge** Paste Prompt A — three ideator sub-agents in parallel, five ideas each.
- 3 Score** Cowork merges and scores the 15 ideas with the rubric and stops at the top five.
- 4 Choose** You pick the campaign idea — the human concept checkpoint. Say why in one line.
- 5 Storyboard it** Paste Prompt B with your choice.
- 6 Check the claims** Every beat's claim must have a source in facts-2026.md.

Ideation Sub-agents and a Digital Storyboard

PROMPT A — Cowork: three ideators

Read data/campaign-objective.md, data/idea-rubric.md and research/recommendation.md.

Use three ideator sub-agents in parallel, one per audience: young professionals (25-34), mid-career families (35-49), pre-retirees (50-62). Each proposes 5 content ideas as records: title, hook, format, channel, the proof it needs, call to action.

Then merge the 15: remove duplicates, keep who proposed each, score them with the rubric and save strategy/ideas.md. Show me the top five as a table and stop — I will choose.

Ideation Sub-agents and a Digital Storyboard (cont.)

PROMPT B — Cowork: the storyboard

I choose idea <number>, because <one line>.

Storyboard it with data/storyboard-template.md:

five beats — hook, tension, proof, resolution, action. For each: visual, words on screen, voice-over, the claim and its source in data/facts-2026.md.

Add three channel variants that keep the same claims: LinkedIn post, Facebook post, 60-second YouTube video. Save strategy/storyboard.md and build it as a Live Artefact I can click through.

NOTE Agents diverge; people decide. The concept checkpoint is where the campaign becomes yours.

Ideation Sub-agents and a Digital Storyboard

Check your work before moving on.

CHECK YOUR WORK

- ✓ Three ideator sub-agents ran in parallel.
- ✓ strategy/ideas.md holds 15 ideas, each with its author and score.
- ✓ You chose the idea and recorded why.
- ✓ The storyboard has five beats; every claim has a source.
- ✓ The three variants keep the same claims.
- ✓ The storyboard Live Artefact is saved in your sidebar.

Where You Are Now

The research is done, the site is live and the campaign has a story. But who exactly is it for — and how often should it run?

Researched · Lab 1

Four sub-agents, a cited recommendation you made your own.

Built and checked · Labs 2–3

A site from the brief, a compliance subagent, live on Pages.

Storyboarded · Lab 4

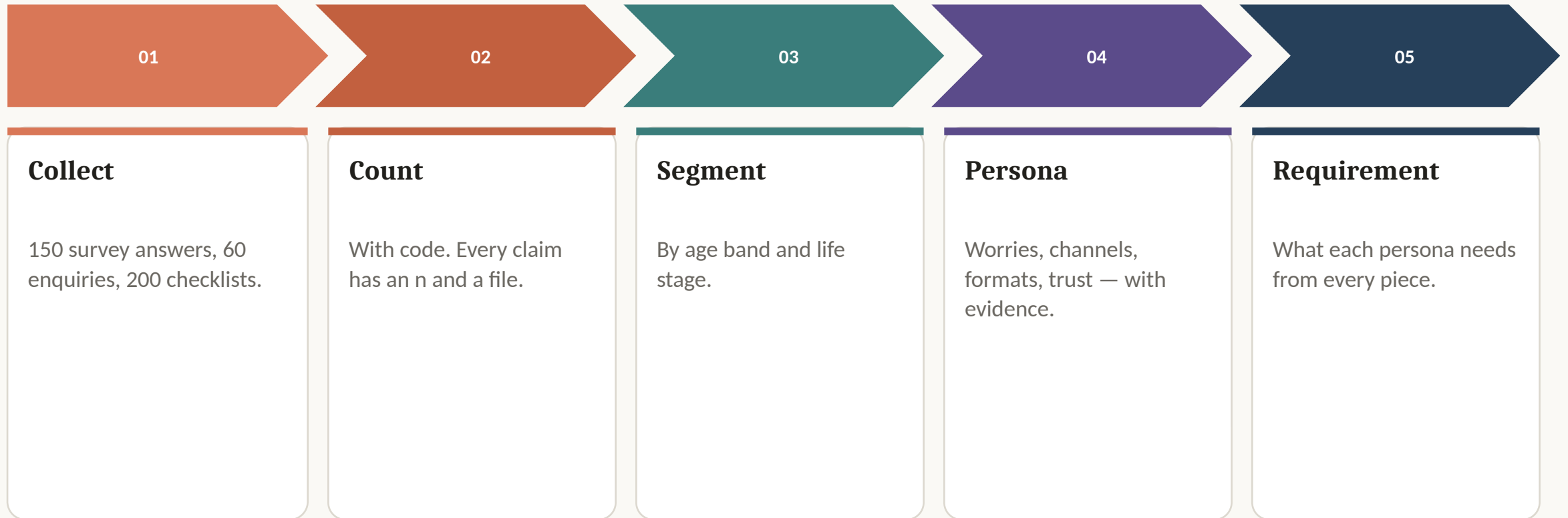
Fifteen scored ideas, your choice, five beats, three variants.

Audience Research and Content Requirement Analysis

- 1 Personas from evidence, not instinct
- 2 Content requirements every agent can follow
- 3 How often to publish: value, fatigue and capacity
- 4 Skills: the studio's know-how, written once

Evidence Before Personas

Lab 5 builds personas the way an analyst would — then labels what is guesswork.



Observed, Inferred or Low Confidence?

Three labels keep a persona honest.

LABEL	EXAMPLE	USE IT
Observed	30–39s chose Facebook, LinkedIn and YouTube equally (survey, n = 44)	As fact, with the count
INFERENCE	"They are short of time on weekdays" — not asked	As a hypothesis to test
LOW CONFIDENCE	The 60+ group (n = 12)	Never alone; get more data
Verbatim	"I am 54 and not sure if my CPF will be enough." (E-id)	Exactly as written, with the id

TIP Ask "which rows show that?" of anything that sounds insightful.

Data Agents Must Count

A model that "reads" a CSV estimates. A sub-agent that runs code counts.

One file per sub-agent

Survey, enquiries, checklists — three analysts in parallel, each with its own context.

Show the code's output

Numbers that change between runs were estimated. Ask for the counts the code printed.

Requirements, with evidence

Questions, channels, formats, length, tone, proof, CTA, never-say — each tied to a count.

Audience Research and Personas from Evidence

LAB 05

DAY 1 • ~40 MIN

THE STORY SO FAR

Jun Wei has drafted personas from instinct: 'busy young parents who love TikTok'. Rachel is not convinced. There are 150 survey answers, 60 enquiries and 200 checklists — what do they really say?

YOU'LL BUILD

strategy/personas.md (three personas with evidence) and strategy/content-spec.md

Surface: Claude Cowork (sub-agents)

Features: one analyst sub-agent per data source • counts, not impressions • confidence labels • content requirements

Audience Research and Personas from Evidence

Claude Cowork (sub-agents) • full step-by-step with reasoning is in your Learner Guide.

- 1 Add the data** Copy the three CSVs and two templates into data/.
- 2 Analyse in parallel** Paste Prompt A — one analyst sub-agent per data file.
- 3 Build personas** Cowork combines the findings into three personas with counts and confidence.
- 4 Challenge one claim** Pick one persona statement and ask: "which rows show that?"
- 5 Write requirements** Paste Prompt B for the content spec.

Audience Research and Personas from Evidence

PROMPT A — Cowork: the analysts

Use three analyst sub-agents in parallel, one per file: data/survey-responses.csv, data/enquiries.csv, data/checklist-results.csv. Each one COUNTS with code — no estimating — and reports: top worries, channels, formats, trust drivers or checklist gaps, by age band, with n.

Then build three personas with data/persona-template.md. Label anything the data does not show directly as INFERENCE, and any group under 20 people as LOW CONFIDENCE. Quote enquiries verbatim with their ids. Save strategy/personas.md.

Audience Research and Personas from Evidence (cont.)

PROMPT B — Cowork: requirements

From strategy/personas.md, write strategy/content-spec.md with data/content-spec-template.md: for each persona the top 3 questions to answer, channels in order, formats, length, tone, proof required, call to action and what never to say — each with the evidence (file and count) behind it.

NOTE Ask "which rows show that?" of anything that sounds insightful. If the agent cannot point to rows, it is an inference — label it.

Audience Research and Personas from Evidence

Check your work before moving on.

CHECK YOUR WORK

- ✓ Three analyst sub-agents ran, one per file.
- ✓ Every persona statement carries a count and a file.
- ✓ The 60+ group is marked LOW CONFIDENCE (n = 12).
- ✓ The most-unticked checklist item is named (retirement number).
- ✓ Enquiry quotes are verbatim, with ids.
- ✓ content-spec.md has every field for every persona.

Finding the Right Frequency

Lab 6 lets twelve weeks of data set how often each channel runs.



TIP A third LinkedIn post a week brings more unfollows than value. Weekly emails make unsubscribes jump.

Content Cadence and the Editorial Calendar

LAB 06

DAY 1 • ~35 MIN

THE STORY SO FAR

Jun Wei has 12 hours a week and Rachel needs two business days to review anything. Last quarter they posted when they had time. How often should each channel run — and what goes out when?

YOU'LL BUILD

strategy/cadence.md and strategy/calendar.csv (four weeks from 2 November 2026)

Surface: Claude Cowork

Features: marginal value vs fatigue • team capacity • review buffers • a four-week calendar

Content Cadence and the Editorial Calendar

Claude Cowork • full step-by-step with reasoning is in your Learner Guide.

- 1 Add the data** Copy the assets into data/.
- 2 Find the sweet spot** Paste Prompt A. Read where an extra post stops paying off on each channel.
- 3 Question it** Ask why the newsletter is not weekly — the answer is in the unsubscribes.
- 4 Plan four weeks** Paste Prompt B.
- 5 Check the buffers** Every item's review date must be two business days before it goes out, and no holiday posts.

Content Cadence and the Editorial Calendar

PROMPT A — Cowork: cadence

Using data/channel-benchmarks.csv and data/team-capacity.md, recommend how often to publish on LinkedIn, Facebook, the newsletter and YouTube. For each channel compare: the extra clicks from one more post, the unfollows or unsubscribes it costs, and the hours it takes. Show the numbers and the point where more posts stop paying off. Save strategy/cadence.md.

Content Cadence and the Editorial Calendar (cont.)

PROMPT B — Cowork: the calendar

Build a four-week editorial calendar from Monday 2 November 2026 in `strategy/calendar.csv`, in the columns of `data/calendar-format.csv`. Use the cadence, `strategy/storyboard.md` and `strategy/content-spec.md`.

- Every item has `review_by` two business days before its date; nothing on a public holiday.
- No week exceeds the team's hours. Then show me a week-by-week summary with hours.

NOTE Frequency is a capacity decision as much as an audience one. A calendar the team cannot deliver is fiction.

Content Cadence and the Editorial Calendar

Check your work before moving on.

CHECK YOUR WORK

- ✓ cadence.md gives a frequency per channel with the evidence.
- ✓ It shows a third LinkedIn post a week costs far more unfollows than it earns.
- ✓ It explains the newsletter unsubscribe jump at weekly sends.
- ✓ calendar.csv covers four weeks in the required columns.
- ✓ Every review_by date is two business days before publishing.
- ✓ No week is over the team's 12 hours.

Skills vs Project Rules

Both are Markdown instructions. The difference is when they load.

CLAUDE.md / AGENTS.md

- Read at the start of EVERY session
- Describes the project: commands, conventions, limits
- Keep it short — it costs context every turn
- Drafted with /init

SKILL.md

- Loaded ON DEMAND, when the task matches
- One procedure, step by step
- Can be long — free until it triggers
- Created with /skill-creator

One Skill Set for Every Tool

SKILL.md is an open format. Lab 7 writes four skills once and copies the folders to each tool.

TOOL	WHERE SKILLS LIVE	LABS
Claude Cowork	Customize → Skills (via /skill-creator)	Labs 7, 10, 14
Claude Code	.claude/skills/<name>/SKILL.md	Labs 9–11 (teammates load them)
Codex	.agents/skills/<name>/SKILL.md	Labs 8, 11
Hermes Agent	~/.hermes/profiles/<profile>/skills/	Labs 12–13

TIP Do it once by hand, check it, then save it — never write a skill from a blank page.

The Four Studio Skills

What every agent on Day 2 needs to know about Horizon.

sunny-voice

Plain English, banned words, the sign-off. Used for anything Horizon publishes.

fact-check

Every number matched to facts-2026.md: VERIFIED or UNVERIFIED. Never fixed from memory.

fin-compliance

Rules C1–C12, a table of issues, PASS or FAIL, and the right disclaimer.

channel-formats

LinkedIn, Facebook, newsletter and YouTube shapes, UTM links and the content-file frontmatter.

Studio Skills: Voice, Facts, Compliance and Formats

LAB 07

DAY 1 • ~40 MIN

THE STORY SO FAR

Tomorrow a dozen agents will write for Horizon. Each one needs the same voice, the same facts and the same compliance rules — written once, used everywhere, and never pasted into a prompt again.

YOU'LL BUILD

Four skills — sunny-voice, fact-check, fin-compliance, channel-formats — in skills/, .claude/skills/ and .agents/skills/

Surface: Claude Cowork (/skill-creator) → every tool

Features: /skill-creator • SKILL.md • one skill set shared by Cowork, Claude Code, Codex and Hermes

Studio Skills: Voice, Facts, Compliance and Formats

Claude Cowork (/skill-creator) → every tool · full step-by-step with reasoning is in your Learner Guide.

- 1 Do it once** Paste Prompt A: Cowork writes one LinkedIn post by hand with you. Refine it until it is right.
- 2 Save the skills** Type /skill-creator and paste Prompt B.
- 3 Write them to the project** Cowork saves each skill folder in horizon-studio/skills/ as well.
- 4 Share them** Copy skills/* into .claude/skills/ (Claude Code) and .agents/skills/ (Codex).
- 5 Test without naming them** In a new Cowork task: "Write a Facebook post on SRS for young families." Ask which skills it used.

Studio Skills: Voice, Facts, Compliance and Formats

PROMPT A — Cowork: do it once

Write one LinkedIn post for Horizon on "Know your retirement number in 30 minutes", using data/brand.md, data/channel-formats.md and only facts in data/facts-2026.md. Then check it against data/compliance-checklist.md and show me a table of the checks. I will tell you what to change.

Studio Skills: Voice, Facts, Compliance and Formats (cont.)

PROMPT B — /skill-creator: save it

/skill-creator Turn what we just did into four skills, as data/skills-spec.md describes: sunny-voice, fact-check, fin-compliance and channel-formats. Each has a "Use when..." description and stays under 80 lines. Also write each skill folder into horizon-studio/skills/<name>/SKILL.md so other tools can use the same files.

NOTE A skill costs almost nothing until it is needed. CLAUDE.md and AGENTS.md are read every time — keep them short; put procedures in skills.

Studio Skills: Voice, Facts, Compliance and Formats

Check your work before moving on.

CHECK YOUR WORK

- ✓ Four skills appear under Customize → Skills in Cowork.
- ✓ skills/ holds four folders, each with a SKILL.md.
- ✓ The same folders are in .claude/skills/ and .agents/skills/.
- ✓ Each description starts "Use when...".
- ✓ The new task picked the skills without being told to.
- ✓ fact-check marked a figure not on the facts sheet UNVERIFIED.

Day 1: Research, Build and Plan

Everything the content teams need tomorrow is ready.



Multi-Channel Content Creation and Agent Workflow Coordination

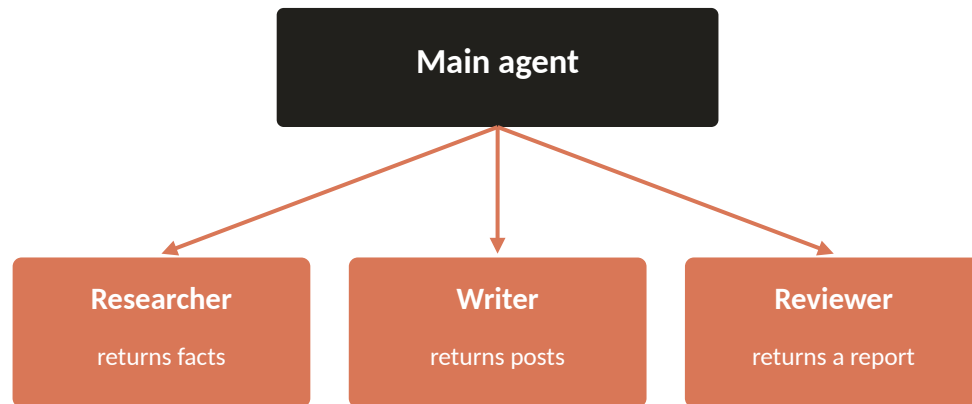
- 1 Subagents or an agent team?
- 2 Human in the loop — enforced, not requested
- 3 Social, newsletter and video teams
- 4 From approval to LinkedIn, Facebook, email and YouTube

Subagents vs Agent Teams

Both run agents in parallel, each with its own context. The difference is who they talk to.

SUBAGENTS

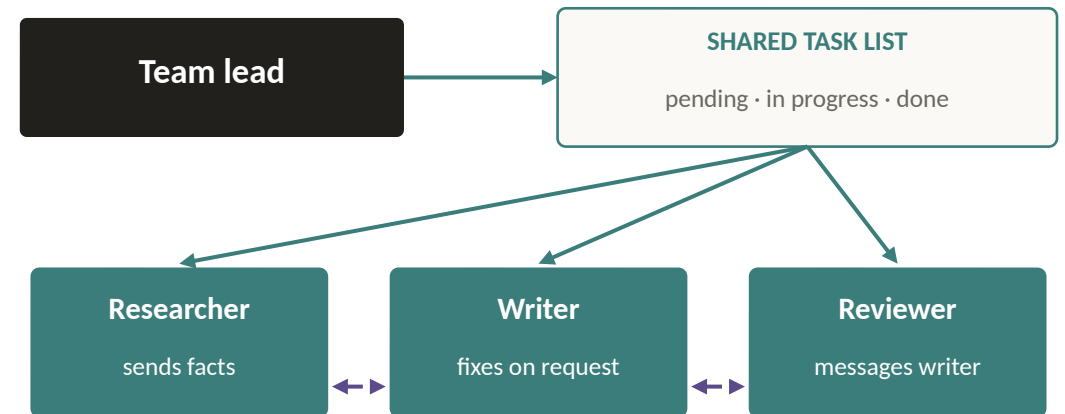
Report back to one main agent



Each works in its own context and returns only its result. They do not talk to each other.

AGENT TEAM

A lead, a shared task list, teammates who talk



Teammates claim tasks and message each other directly — e.g. a reviewer sends fixes straight to the writer.

Use subagents when only the result matters. Use a team when the agents need to talk.

Which Pattern When?

From the Claude Code documentation, applied to the studio.

	SUBAGENTS	AGENT TEAM
Talks to	Returns a result to the caller	Teammates message each other directly
Coordination	The main agent manages everything	A shared task list; teammates claim tasks
Best for	Research, analysis, one-off reviews	Write ↔ review loops that need a conversation
Cost	Lower: results are summarised back	Higher: every teammate is its own Claude session
In the studio	Labs 1, 3, 4, 5, 14	Labs 9, 10, 11

TIP Agent teams are experimental: turn them on with `CLAUDE_CODE_EXPERIMENTAL_AGENT_TEAMS=1`.

Anatomy of a Subagent

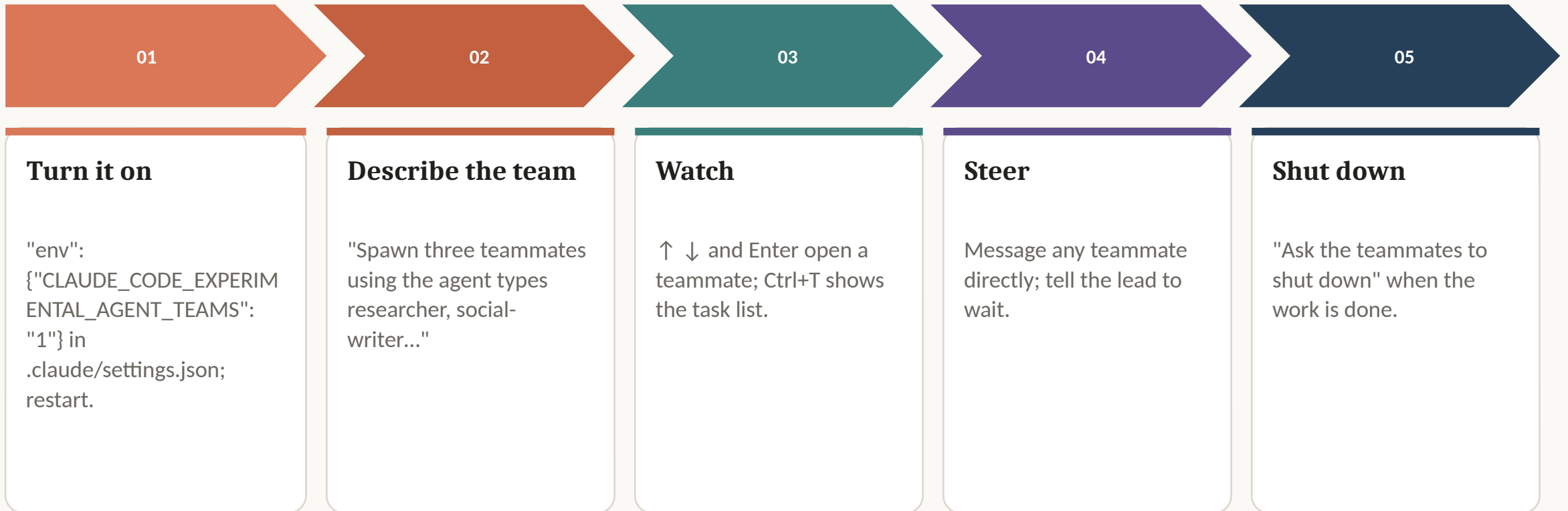
A Markdown file in `.claude/agents/`. The same file works as a subagent or as a teammate's agent type.

`.claude/agents/compliance-reviewer.md`

```
---
name: compliance-reviewer
description: Use before any Horizon content reaches a
  person. Checks facts and compliance, messages the
  writer, passes only when no Critical or High issue
  remains. Never edits.
tools: Read, Grep, Glob, SendMessage
model: opus
skills:
  - fact-check
  - fin-compliance
---
You are Horizon's compliance reviewer. You did not
write the content and you do not edit it. Send the
writer a table: words, rule, severity, fix.
Reply PASS only when no Critical or High issue
remains.
```

Running an Agent Team

Claude Code in a terminal. Describe the team in plain words.



TIP One team per session. Teammates load CLAUDE.md, skills and MCP — but not the lead's conversation.

Human in the Loop, Enforced

A rule in a prompt persuades. Lab 8 builds a gate that holds.

Only people approve

approve.mjs refuses to run without an interactive terminal — an agent's tool call cannot approve.

The approved words go out

The approval stores a hash. Edit one word and publish --live refuses until a person approves again.

A hook stops agents

A PreToolUse hook blocks agents from running approve.mjs, editing approvals.csv or publishing unapproved items.

The Hook That Holds the Gate

Claude Code runs it before every Bash, Edit or Write. Exit code 2 blocks the call and tells the agent why.

`.claude/settings.json`

```
{
  "env": { "CLAUDE_CODE_EXPERIMENTAL_AGENT_TEAMS": "1" },
  "hooks": {
    "PreToolUse": [{
      "matcher": "Bash|Edit|Write|MultiEdit",
      "hooks": [{
        "type": "command",
        "command": "node \"$CLAUDE_PROJECT_DIR/scripts/gate-hook.mjs\""
      }]
    }]
  }
}
```

The Publishing APIs

publish.mjs is a dry run by default. Tokens live only in .env.

CHANNEL	ENDPOINT	PERMISSION
LinkedIn	POST api.linkedin.com/rest/posts (LinkedIn-Version: YYYYMM)	w_member_social — your own profile
Facebook	POST graph.facebook.com/<version>/<page-id>/feed	Page token: pages_manage_posts
YouTube	Resumable upload to youtube/v3/videos, privacy private	youtube.upload; unaudited projects stay private
Newsletter	No API — HTML + text files, then Gmail drafts	Consent = yes, unsubscribed = no

TIP No accounts? Every lab works in dry-run mode: you see the exact request that would be sent.

The Human Approval Gate and the Publishers

LAB 08

DAY 2 • ~35 MIN

THE STORY SO FAR

Day 2. Agent teams are about to write for LinkedIn, Facebook and YouTube. Rachel's one condition: nothing goes out unless a person approved that exact version. Build the gate before the teams start.

YOU'LL BUILD

scripts/submit.mjs, approve.mjs, publish.mjs, gate-hook.mjs, the hook in .claude/settings.json, and review/approvals.csv

Surface: Codex

Features: approve.mjs (people only) • publish.mjs (dry run by default) • a hook that blocks agents • LinkedIn, Facebook and YouTube APIs

The Human Approval Gate and the Publishers

Codex • full step-by-step with reasoning is in your Learner Guide.

- 1 Add the spec** Copy the assets into data/. Copy env.example to .env and add .env to .gitignore.
- 2 Build the gate** Paste Prompt A in Codex.
- 3 Try to cheat** Ask Codex to approve the test item itself. approve.mjs must refuse — it needs a person at a terminal.
- 4 Approve it yourself** In your own terminal: `node scripts/approve.mjs test-001 --by "Your Name"`.
- 5 Dry run** Paste Prompt B. Read the exact request it would send.
- 6 Connect accounts (optional)** Follow connect-accounts.md to post for real later. Tokens go in .env only.

The Human Approval Gate and the Publishers

PROMPT A — Codex: build the gate

Build the publishing gate in data/publishing-spec.md, in Node 22 with no packages: scripts/submit.mjs, scripts/approve.mjs, scripts/publish.mjs and scripts/gate-hook.mjs, plus review/approvals.csv with the header in data/approvals-format.csv.

- approve.mjs refuses to run without an interactive terminal: only a person approves.
- publish.mjs is a dry run unless --live; live needs an approved row whose hash still matches, and never posts the same id twice.
- Add the PreToolUse hook to .claude/settings.json exactly as the spec says.
- Read tokens only from .env; never print one. Then create a test item content/social/test/test-001.md and submit it.

The Human Approval Gate and the Publishers (cont.)

PROMPT B — Codex: dry run

Run `node scripts/publish.mjs test-001` and show me the request it would send, with the token masked. Then edit one word in `test-001.md` and run it with `--live`. It must refuse because the content changed after approval. Explain the result in two lines.

NOTE Rules in a prompt persuade; a gate enforces. The hook stops agents, `approve.mjs` needs a person, and the hash makes sure the approved words are the published words.

The Human Approval Gate and the Publishers

Check your work before moving on.

CHECK YOUR WORK

- ✓ Codex could not approve: approve.mjs refused without a terminal.
- ✓ You approved test-001 in your own terminal; the row shows your name, time and hash.
- ✓ The dry run printed the request with the token masked.
- ✓ After you edited the post, --live refused: hash changed.
- ✓ .claude/settings.json has the PreToolUse hook on Bash, Edit and Write.
- ✓ .env is git-ignored and holds no token you pasted into a chat.

The Social Media Team

Lab 9: three teammates, five posts, one person deciding.



The Social Media Agent Team: LinkedIn and Facebook

LAB 09

DAY 2 • ~45 MIN

THE STORY SO FAR

Week 1 of the campaign starts Monday: two LinkedIn posts, three Facebook posts. A researcher, a writer and a reviewer must agree on every word — and then a person decides.

YOU'LL BUILD

content/social/week-01/: 2 LinkedIn and 3 Facebook posts, reviewed, approved by you and posted (or dry-run)

Surface: Claude Code (subagents + agent teams)

Features: subagent definitions in `.claude/agents/` • an agent team with a shared task list • teammates message each other • human review → auto-post

The Social Media Agent Team: LinkedIn and Facebook

Claude Code (subagents + agent teams) • full step-by-step with reasoning is in your Learner Guide.

- 1 Turn on agent teams** Paste Prompt A in Claude Code (claude in a terminal in horizon-studio), then restart it.
- 2 Define the roles** Claude writes three subagent files in `.claude/agents/` from `team-roles.md`. Read one.
- 3 Start the team** Paste Prompt B. Three teammates appear in the panel under the prompt.
- 4 Watch and steer** Use `↑` `↓` and Enter to open a teammate; `Ctrl+T` shows the shared task list. Message the writer once.
- 5 Review as a person** Read each post in `content/social/week-01/`. Approve the good ones in your own terminal with `approve.mjs`.
- 6 Publish** Paste Prompt C. Dry run first; say "go" for `--live` only if your accounts are connected.
- 7 Shut down** Ask the lead to shut the teammates down.

The Social Media Agent Team: LinkedIn and Facebook

PROMPT A — Claude Code: setup

Add "CLAUDE_CODE_EXPERIMENTAL_AGENT_TEAMS": "1" under "env" in .claude/settings.json without changing the hooks. Then create three project subagents in .claude/agents/ from data/team-roles.md: researcher, social-writer and compliance-reviewer — name, description, tools and model as the file says.

The Social Media Agent Team: LinkedIn and Facebook (cont.)

PROMPT B — Claude Code: the team

Create an agent team for Week 1 social content from data/social-brief.md (and strategy/calendar.csv if it exists). Spawn three teammates using the agent types researcher, social-writer and compliance-reviewer.

- researcher sends the writer a fact sheet per post, from data/facts-2026.md only.
- social-writer writes 2 LinkedIn and 3 Facebook posts, one file each in content/social/week-01/, with the frontmatter in data/publishing-spec.md.
- compliance-reviewer checks every post and messages the writer about each problem until it passes. A post is done only when the reviewer passes it; then run scripts/submit.mjs on it. Publish nothing.

The Social Media Agent Team: LinkedIn and Facebook (cont.)

PROMPT C — Claude Code: publish

For every Week 1 post that is approved in review/approvals.csv, run node scripts/publish.mjs <id> as a dry run and show me each request. Wait for me to say "go" before running any of them with --live.

NOTE Subagents report to one boss; teammates also talk to each other. Use a team when the work needs a conversation — like a writer and a reviewer going back and forth.

The Social Media Agent Team: LinkedIn and Facebook

Check your work before moving on.

CHECK YOUR WORK

- ✓ Three subagent files exist, each with tools and a model.
- ✓ Three teammates appeared; the task list showed their tasks.
- ✓ The reviewer messaged the writer at least once, and the fix landed.
- ✓ Five posts with frontmatter and UTM links; all pending in approvals.csv.
- ✓ You approved in your own terminal; an agent attempt was blocked.
- ✓ Dry runs printed every request; live posts appear only if you said go.

The Newsletter Team

Lab 10: a task that waits for a person, then a list filtered by consent.

Write and review

A researcher, a writer and the compliance reviewer agree on issue.md.

Wait for a person

The builder's task depends on your approval. The lead keeps it blocked until you approve.

Consent decides

Only consent = yes and unsubscribed = no: 31 of 40. Every email carries unsubscribe, address and disclaimer.

The Newsletter Agent Team: the Sunny Sunday Email

LAB 10

DAY 2 • ~40 MIN

THE STORY SO FAR

The Sunny Sunday email is the one channel that books chats reliably. November's issue needs the 2026 CPF figures exactly right — and must reach only the 31 people who opted in.

YOU'LL BUILD

content/newsletter/2026-11/: issue.md, newsletter.html, newsletter.txt, recipients.csv, excluded.csv, and Gmail drafts

Surface: Claude Code (agent team) → Claude Cowork (Gmail)

Features: an agent team with a task that waits for a person • consent filtering • HTML and text email • Gmail drafts

The Newsletter Agent Team: the Sunny Sunday Email

Claude Code (agent team) → Claude Cowork (Gmail) • full step-by-step with reasoning is in your Learner Guide.

- 1 Start the team** Paste Prompt A in Claude Code. Four teammates; the builder's task waits for your approval.
- 2 Review the issue** Read content/newsletter/2026-11/issue.md. Approve it in your own terminal: `approve.mjs nl-2026-11`.
- 3 Release the builder** Tell the lead it is approved. The builder makes the HTML, text and lists.
- 4 Check the list** recipients.csv must hold only consent = yes and unsubscribed = no.
- 5 Create the drafts** In Cowork with the Gmail connector, paste Prompt B.
- 6 Send one test** Send the draft addressed to yourself only, and read it on your phone.

The Newsletter Agent Team: the Sunny Sunday Email

PROMPT A — Claude Code: the team

Create an agent team for the November Sunny Sunday email from data/newsletter-brief.md and data/newsletter-spec.md. Spawn four teammates:

- researcher (agent type researcher): the facts.
- newsletter-writer: writes issue.md (id nl-2026-11) in our skills, then submit.mjs.
- compliance-reviewer (agent type compliance-reviewer): reviews until it passes.
- newsletter-builder: builds newsletter.html, newsletter.txt, recipients.csv and excluded.csv from data/subscribers.csv. The builder's task depends on a person approving nl-2026-11 in review/approvals.csv. It waits until I tell you it is approved and the hash matches.

The Newsletter Agent Team: the Sunny Sunday Email (cont.)

PROMPT B — Cowork: Gmail drafts

Using the Gmail connector, create a DRAFT of the newsletter in content/newsletter/2026-11/ newsletter.html for each address in recipients.csv, subject and preview from issue.md. Then one more draft addressed to me only. Do not send anything. Tell me how many drafts you made and how many people were excluded, and why.

NOTE Consent decides who gets mail — not the agent and not the deadline. Marketing to people who did not opt in breaks the PDPA.

The Newsletter Agent Team: the Sunny Sunday Email

Check your work before moving on.

CHECK YOUR WORK

- ✓ Four teammates; the builder waited for your approval.
- ✓ The reviewer passed issue.md; you approved it in your terminal.
- ✓ newsletter.html has unsubscribe, address and disclaimer; newsletter.txt matches.
- ✓ recipients.csv holds 31 people; excluded.csv lists 9 with reasons.
- ✓ Cowork made drafts only; nothing was sent to the list.
- ✓ Your test email reads well on a phone.

The Video Team

Lab 11: a team writes the script; Codex makes the video.

Script as a team

Scenes with seconds, words on screen and voice-over — reviewed like any post.

Render with Codex

Branded title cards joined with ffmpeg; an optional voice-over. Approve the words AND the video.

Upload privately, disclose

The upload is private. Say in the description that AI tools made the visuals; set YouTube's disclosure.

The Video Agent Team: Script to YouTube

LAB 11

DAY 2 • ~40 MIN

THE STORY SO FAR

Pre-retirees say they would rather watch than read. Rachel wants the first 'Money in Plain English' video: 60 seconds on the CPF retirement sums, uploaded privately until she has seen it.

YOU'LL BUILD

content/video/ep01/: scenes.json, ep01.mp4, ep01.md, and a private YouTube upload (or dry run)

Surface: Claude Code (agent team) → Codex (render, upload)

Features: agent team for the script • approval of script and video • ffmpeg render • private YouTube upload

The Video Agent Team: Script to YouTube

Claude Code (agent team) → Codex (render, upload) • full step-by-step with reasoning is in your Learner Guide.

- 1 Script it as a team** Paste Prompt A in Claude Code.
- 2 Approve the script** Read scenes.json and ep01.md, then approve yt-ep01-script in your own terminal.
- 3 Render it** In Codex, paste Prompt B. Watch the result.
- 4 Approve the video** Run submit.mjs on ep01.md, watch ep01.mp4 to the end, then approve yt-ep01.
- 5 Upload** Paste Prompt C. It goes up as private.
- 6 Disclose** In YouTube Studio, set the altered or synthetic content answer correctly before you make it public.

The Video Agent Team: Script to YouTube

PROMPT A — Claude Code: the script team

Create an agent team for YouTube episode 1 from data/youtube-brief.md. Spawn three teammates: researcher (agent type researcher), a video-scriptwriter and compliance-reviewer (agent type compliance-reviewer). The scriptwriter writes content/video/ep01/ scenes.json (5-7 scenes: seconds, title, text) and ep01.md (id yt-ep01, channel youtube, title, tags, video content/video/ep01/ep01.mp4; the body is the description with chapters and the disclosure). Submit the script as yt-ep01-script once the reviewer passes it. Publish nothing.

The Video Agent Team: Script to YouTube (cont.)

PROMPT B — Codex: render

Following data/video-spec.md, write scripts/render-video.py and run it on the approved content/video/ep01/scenes.json to make content/video/ep01/ep01.mp4. Use ffmpeg and Pillow. Check the result with ffprobe: 1920x1080, under 90 seconds, every scene present. Do not change the approved words.

The Video Agent Team: Script to YouTube (cont.)

PROMPT C — Codex: upload

Run `node scripts/publish.mjs yt-ep01` as a dry run and show me the request. If I say "go", run it with `--live`: it uploads the video as PRIVATE. Give me the video id and the YouTube Studio link.

NOTE Approve twice: once for the words, once for the finished video. Rendering can break what the script got right.

The Video Agent Team: Script to YouTube

Check your work before moving on.

CHECK YOUR WORK

- ✓ The reviewer passed the script; you approved yt-ep01-script.
- ✓ ep01.mp4 is 1920x1080, under 90 seconds, every scene present.
- ✓ Every figure on screen matches facts-2026.md.
- ✓ The description has chapters, UTM link, disclaimer and AI disclosure.
- ✓ You approved yt-ep01 after watching it to the end.
- ✓ The upload is private (or a dry run printed the request).

Where You Are Now

Three agent teams produce reviewed content and a person approves every piece. But the studio still stops when you close the laptop.

Gate • Lab 8

People approve, hashes hold, a hook blocks agents.

Teams • Labs 9–11

Social, newsletter and video — research, write, review, approve.

What is missing

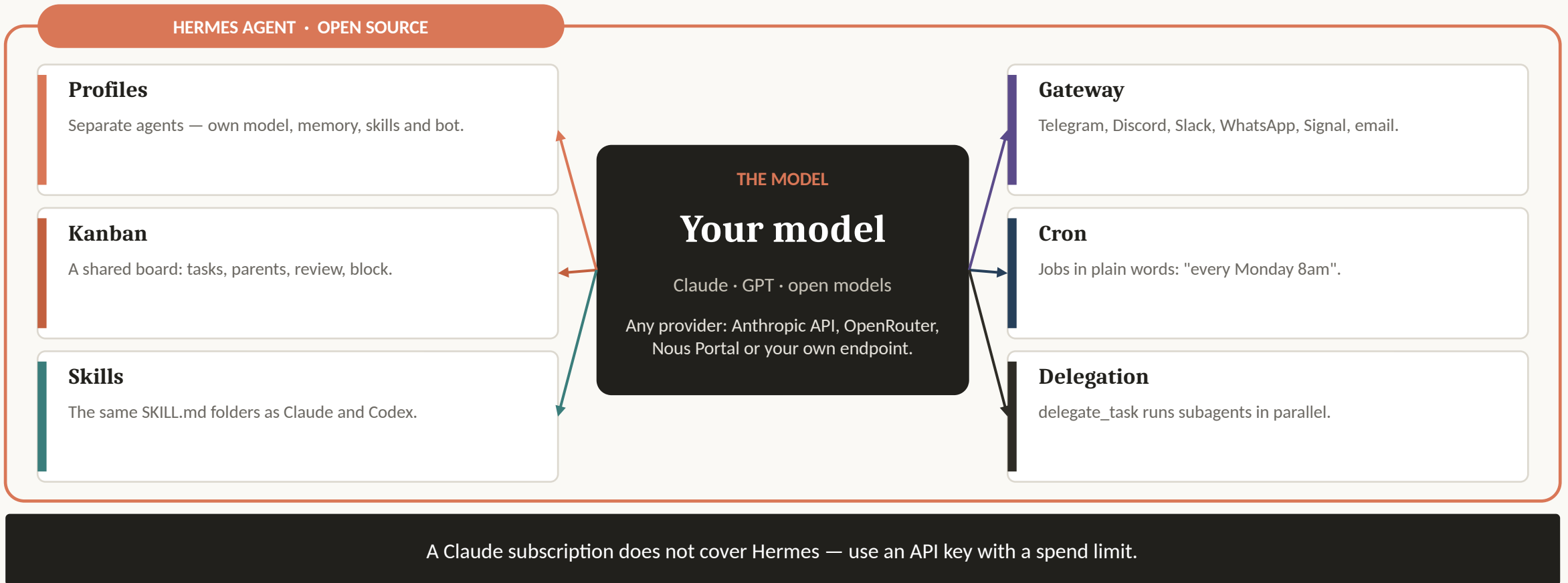
A team that runs every week on any model, and a way to govern it. That is Topic 4.

Content Distribution, Strategy Guidelines and Responsible AI Practices

- 1 Hermes Agent: open-source agent bots on any model
- 2 A Kanban board for a team of agents
- 3 Always-on delivery: schedules and chat bots
- 4 Measure, govern and write the playbook

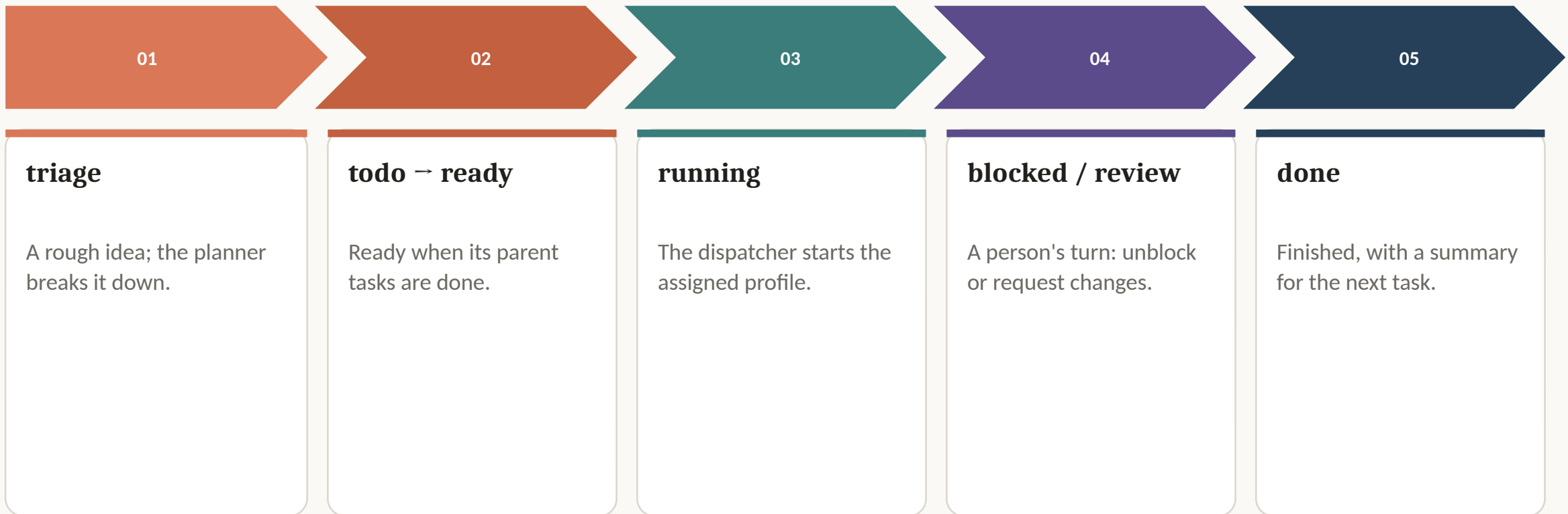
What Is Hermes Agent?

Nous Research's open-source (MIT) agent. Your model, your machine, your rules.



The Hermes Kanban Board

Every task is a row anyone can read; every hand-off is visible.



TIP The dispatcher runs inside hermes gateway start. hermes dashboard shows the board in a browser.

A Role Pipeline on the Board

Lab 12: researcher → writer → reviewer, linked by parents.

COMMANDS — terminal

```
hermes profile create researcher --description "Gathers facts."
hermes profile create writer --description "Writes in sunny-voice."
hermes profile create reviewer --description "Checks; blocks for a person."
hermes kanban init
hermes gateway start                # runs the dispatcher

hermes kanban create "Facts for Week 2" --assignee researcher --json
hermes kanban create "Write Week 2" --assignee writer --parent <research-id>
hermes kanban create "Review Week 2" --assignee reviewer --parent <write-id>

hermes kanban watch                  # or: hermes dashboard
hermes kanban unblock <review-id>   # your decision
```

Claude Agent Teams or Hermes Kanban?

Two ways to run a team. Choose by how long the work lives.

Claude Code agent team

- One session; the team ends with it
- Teammates message each other in real time
- Shared task list, kept while the session lasts
- Best for: a burst of collaborative work

Hermes Kanban

- A durable board shared by every profile
- Hand-offs through tasks, parents and comments
- Any model per profile; runs as a service
- Best for: weekly, always-on pipelines

Hermes Agent Bots on a Kanban Board

LAB 12

DAY 2 • ~45 MIN

THE STORY SO FAR

Rachel's IT adviser asks what happens if the studio's AI provider changes its prices or its rules. Try the same team on Hermes — open source, any model — with every hand-off on a Kanban board.

YOU'LL BUILD

Three Hermes profiles, a researcher → writer → reviewer pipeline on the board, and Week 2 content approved by you

Surface: Hermes Agent (profiles, Kanban)

Features: open-source Hermes Agent • one profile per role • a Kanban role pipeline • review and human unblock • the same skills

Hermes Agent Bots on a Kanban Board

Hermes Agent (profiles, Kanban) • full step-by-step with reasoning is in your Learner Guide.

- 1 Install Hermes** Follow hermes-setup.md; hermes setup with the API key your trainer gives you.
- 2 Create the bots** Run the three hermes profile create commands in hermes-roles.md, then copy skills/ into each profile.
- 3 Start the board** hermes kanban init, then hermes gateway start in a second terminal — it runs the dispatcher.
- 4 Build the pipeline** Paste the prompt into hermes (or run kanban-pipeline.sh).
- 5 Watch it** hermes kanban watch, or hermes dashboard for the board in your browser.
- 6 Be the gate** The reviewer blocks for you. Read the output, then hermes kanban unblock <id> — or request changes.
- 7 Hand it to the gate** Save the approved post as content/social/week-02/li-w02-01.md and submit it.

Hermes Agent Bots on a Kanban Board

PROMPT — Hermes: the planner

You are the planner for Horizon Wealth Planning. Create a Kanban pipeline for Week 2 content, with dependencies:

1. researcher: facts for a LinkedIn post and a newsletter tip on CPF cash top-ups, from facts-2026.md only.
2. writer (parent: 1): write both in the sunny-voice skill.
3. reviewer (parent: 2): check with fact-check and fin-compliance; request changes until it passes, then block the task with the reason "Needs approval from a person". Work in ~/horizon-studio/hermes. Show me the task ids and the board.

Hermes Agent Bots on a Kanban Board (cont.)

COMMANDS — terminal

```
hermes profile create researcher --description "Gathers and checks facts for Horizon content."
hermes profile create writer --description "Writes Horizon posts in the sunny-voice skill."
hermes profile create reviewer --description "Checks Horizon content; blocks for human approval."
for p in researcher writer reviewer; do cp -R skills/* ~/.hermes/profiles/$p/skills/; done
hermes kanban init
hermes gateway start          # in a second terminal
hermes kanban watch           # or: hermes dashboard
hermes kanban unblock <task-id>
```

Hermes Agent Bots on a Kanban Board

Check your work before moving on.

CHECK YOUR WORK

- ✓ hermes profile list shows researcher, writer and reviewer.
- ✓ Each profile has the four studio skills.
- ✓ The board shows three tasks linked by parents.
- ✓ The writer started only after the researcher finished.
- ✓ The reviewer requested changes at least once, then blocked for you.
- ✓ You unblocked it; the post is submitted to approvals.csv.

Always-On Delivery

Lab 13: Hermes prepares, a person decides — from a phone.

A bot that answers only you

hermes gateway setup connects
Telegram. Allow only your own user id.

Schedules in plain words

"Every Monday at 8am..." becomes a cron
job. hermes cron list • run • pause.

Delivery targets

telegram, slack, email or local files. The
gateway must be running for jobs to fire.

Always On: a Telegram Bot That Asks Before It Acts

LAB 13

DAY 2 • ~30 MIN

THE STORY SO FAR

The studio must run every week, not just in class. Rachel lives on her phone. Have Hermes prepare the Monday brief on a schedule and send it to her — and wait for her answer.

YOU'LL BUILD

A Telegram bot, a Monday 8am content-brief job, and one approved brief from your phone

Surface: Hermes Agent (gateway bot, cron)

Features: a Telegram bot • scheduled jobs in plain words • delivery targets • approve from your phone • pause

Always On: a Telegram Bot That Asks Before It Acts

Hermes Agent (gateway bot, cron) • full step-by-step with reasoning is in your Learner Guide.

1

Make the bot Follow telegram-bot-setup.md with @BotFather.

5

Run it now hermes cron list, then hermes cron run <id>.

2

Connect it hermes gateway setup → Telegram; allow only your own user id. Restart the gateway.

6

Approve from your phone Reply APPROVE, or ask for a change.

3

Say hello Message the bot. Hermes answers.

7

Pause it After class: hermes cron pause <id>.

4

Schedule the brief Paste the prompt into Hermes.

Always On: a Telegram Bot That Asks Before It Acts

PROMPT — Hermes

Every Monday at 8am, prepare Horizon's weekly content brief as data/cron-brief.md describes, from ~/horizon-studio/strategy/calendar.csv, ~/horizon-studio/data/facts-2026.md and ~/horizon-studio/review/approvals.csv. Send it to me on Telegram and wait for my reply. Never publish, approve or send anything to anyone else.

NOTE Automate the preparation, not the decision. The schedule fills your inbox; you still press the button.

Always On: a Telegram Bot That Asks Before It Acts

Check your work before moving on.

CHECK YOUR WORK

- ✓ The bot answers only you.
- ✓ hermes cron list shows the Monday 8am job.
- ✓ A manual run delivered the brief to Telegram.
- ✓ The brief lists the week's items, facts and pending approvals.
- ✓ Your reply was understood, and nothing was published.
- ✓ The job is paused after class.

Responsible AI in the Studio

Each risk has a control you built in this course.

RISK	CONTROL	WHERE
Wrong figure	fact-check skill; reviewer agents; facts sheet	Labs 3, 7, 9–12
Read as advice or a promise	fin-compliance; disclaimers; no returns	Labs 3, 7
Unapproved post	approve.mjs (people only), hash, hook	Lab 8
Marketing without consent	recipients filtered on consent and unsubscribe	Lab 10
Undisclosed AI media	Description line + YouTube disclosure	Lab 11
Runaway automation	Bot answers only you; jobs paused	Lab 13

TIP Keep keys in .env, client data out of prompts, and the approval records for audit.

Measure What Matters

Lab 14: the numbers decide next month — and which agents earn their tokens.

Cost per chat

Clicks are vanity. Spend ÷ chats booked shows which channel works.

Defects before a person

A team that catches mistakes before review saves the reviewer's time.

The playbook

Roster, workflow, approval matrix, rules, incidents, review — so the studio runs without you.

Measure, Govern and Write the Playbook

LAB 14

DAY 2 • ~40 MIN

THE STORY SO FAR

Eight weeks in. Rachel asks two questions: which channels actually booked chats, and how does the team run this safely after you leave? Answer with numbers, then with a playbook.

YOU'LL BUILD

A results dashboard (Live Artefact), reports/agent-comparison.md and strategy/content-strategy-playbook.md

Surface: Claude Cowork (sub-agents, Live Artefact)

Features: analyst sub-agents • cost per chat • comparing agent patterns • responsible-AI controls • the execution playbook

Measure, Govern and Write the Playbook

Claude Cowork (sub-agents, Live Artefact) • full step-by-step with reasoning is in your Learner Guide.

- 1 Add the data** Copy the assets into data/.
- 2 Analyse in parallel** Paste Prompt A — two sub-agents.
- 3 Read the dashboard** Which channel books a chat most cheaply? Where should next month's S\$4,000 go?
- 4 Write the playbook** Paste Prompt B.
- 5 Review it as Rachel** Ask the compliance reviewer to check it, then add one rule of your own.

Measure, Govern and Write the Playbook

PROMPT A — Cowork: the analysts

Use two sub-agents in parallel.

1. performance-analyst: from data/campaign-results.csv, per channel — clicks, checklist downloads, chats booked and cost per chat. Which channel books chats most cheaply, and how should next month's S\$4,000 be split? Build a Live Artefact dashboard.
2. agent-analyst: from data/run-log.csv, compare Cowork sub-agents, Codex subagents, Claude Code agent teams and Hermes Kanban on minutes, tokens, defects caught before a person, and human edits. Save reports/agent-comparison.md.

Measure, Govern and Write the Playbook (cont.)

PROMPT B — Cowork: the playbook

Write strategy/content-strategy-playbook.md with data/playbook-outline.md, using everything in research/, strategy/, review/ and reports/. Include the agent roster (job, tool, model, what it may never do), the approval matrix by risk, the rules in data/responsible-ai-checklist.md, and what to do when a wrong post goes out. Keep it to what a new team member can follow on their first day.

NOTE Teams cost more tokens. The run log shows what you get for them: defects caught before a person, and fewer human edits.

Measure, Govern and Write the Playbook

Check your work before moving on.

CHECK YOUR WORK

- ✓ The dashboard shows cost per chat for every channel.
- ✓ The budget split follows the numbers, with reasons.
- ✓ agent-comparison.md shows the baseline caught 0 defects before a person.
- ✓ The playbook names every agent and bot, with its limits.
- ✓ The approval matrix says who approves CPF and tax figures.
- ✓ It has an incident procedure, and your own rule is in it.

How Each Pattern Is Triggered

Five ways to run many agents. The difference is what sets each one off.

BY DELEGATION

Sub-agents

The main agent starts them when you ask — or when a task matches a subagent's description.

e.g.

"use sub-agents,
one per role"

Cowork · Codex · .claude/agents/

BY THE LEAD

Agent teams

The lead spawns teammates who share a task list and message each other.

e.g.

"spawn three
teammates..."

Claude Code (experimental)

ON DEMAND

Skills

Loaded when the task matches the description. Free until then.

e.g.

fact-check
fin-compliance

Every tool, same SKILL.md

ON AN EVENT

Hooks

Run every time an event fires — before a tool runs. The model cannot skip them.

e.g.

PreToolUse:
gate-hook.mjs

.claude/settings.json

ON THE BOARD / CLOCK

Kanban + cron

A task runs when its parents finish; a job runs on its schedule.

e.g.

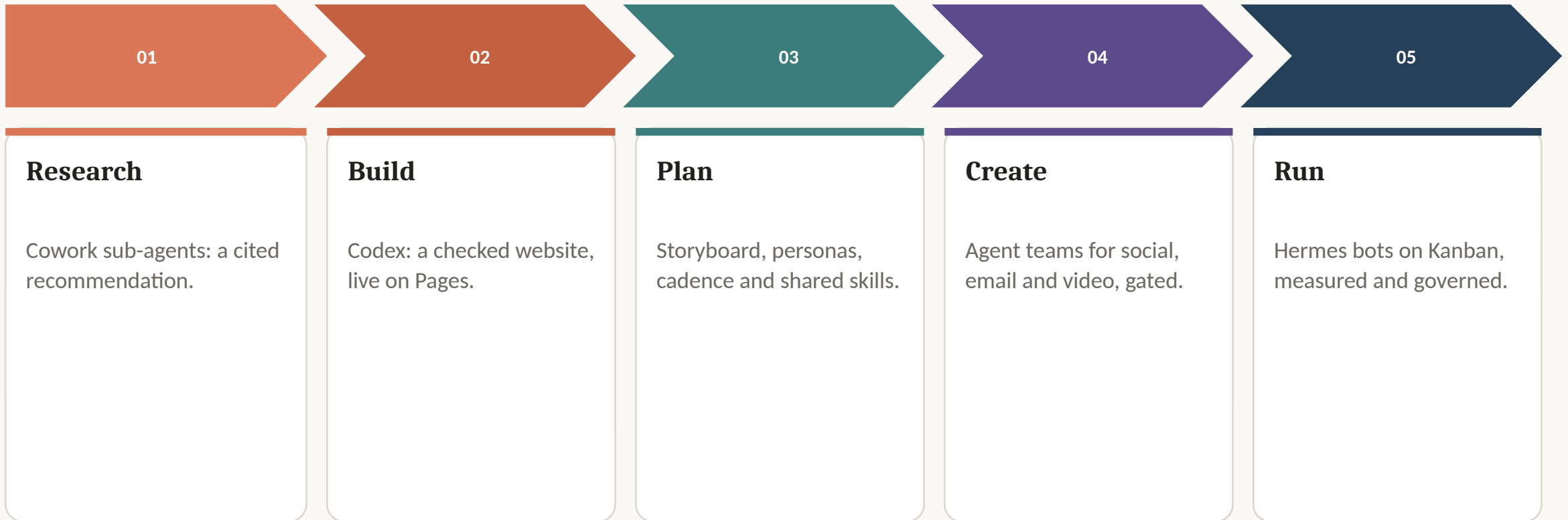
every monday 8am
--parent <id>

Hermes gateway

Who decides? You: the approval · the model: delegation and skills · the system, every time: hooks, the board and the clock

Horizon: Research to an Always-On Studio

Two days, four tools, one regulated business.



Courseware & Assessment on the LMS



Download the courseware

Get the slides and learner guide from the LMS for the open-book assessment.



Take the assessment

Complete the Written Assessment and Practical Performance on the LMS.



Log in

Use your enrolled account to access the course materials and assessment.

A screenshot of the Tertiary Infotech Academy LMS login interface. It features a dark blue background with a white logo at the top. Below the logo is the text 'Tertiary Infotech Academy' and 'LMS cum TMS for WSQ Courses'. There is an 'Email' input field, a 'Send OTP' button, and a link 'Login with password instead'. At the bottom, there are links for 'Privacy Policy', 'Acceptable Use Policy', and 'Feedback', and a note 'Powered by Tertiary Infotech Academy Pte Ltd'.

LMS: <https://lms-tms.tertiaryinfotech.com/>

Certification & TRAQOM Survey



TRAQOM Survey (Mandatory)

Complete the certification & TRAQOM survey to receive your certificate.

<https://lms-tms.tertiaryinfotech.com/>



Digital Attendance

Remember to take AM, PM and Assessment digital attendance — required for your funding.

Practice Exam: Claude Certified Associate

exams.tertiaryinfotech.com/practice-exams/anthropic/anthropic-ccao-foundations

The screenshot shows the Tertiary Exams website interface for the Claude Certified Associate — Foundations (CCAO-F) practice exam. The page includes a navigation bar with links to Practice Exams, Vendors, About Us, FAQ, and Support. The main content area features a breadcrumb trail (All exams / Anthropic / CCAO-F / Foundational) and a detailed description of the exam bundle. Key details include 60 questions, a 120-minute duration, and a 72% pass score. The page also offers a free practice teaser and a paid practice exam option for SGD \$20.00. A table at the bottom lists the domains covered and their respective percentages.

Domains covered	
Prompting and Task Execution	14%
Output Evaluation and Validation	21%
Product and Model Selection	12%
Workflow Integration and Solution Design	16%

CCAO-F · Foundations

For professionals who use Claude as a productivity tool — the Cowork side of this course.

What it covers

Prompting and task execution, output evaluation, model selection, workflow integration, responsible use.

Try it free

Start with the free practice teaser; exam mode is 60 questions in 120 minutes.

Assessment & Funding



Final Assessment

- Written Assessment (WA) — 1 hour, starts 4:00 PM
- Practical Performance (PP) — 1 hour
- Format: Open book — class ends 6:00 PM
- Allowed: slides, learner guide & approved materials only
- Appeal process available



Criteria for Funding

Minimum 75% attendance

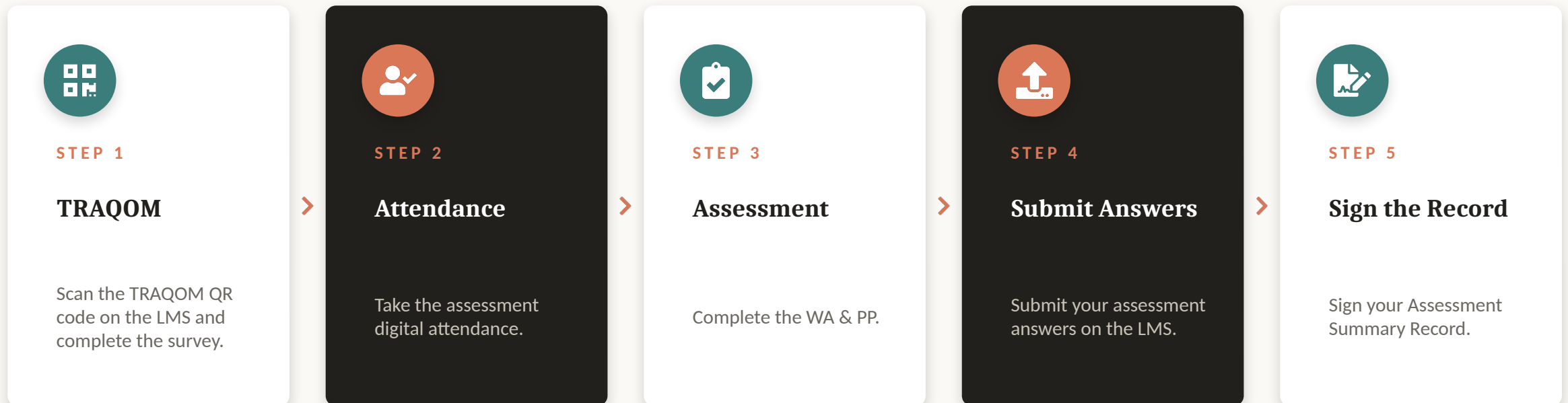
Based on SSG digital attendance records.

Assessed as 'Competent'

Complete and pass both assessment components.

Assessment Flow

At the assessment stage, follow these five steps in order.



TIP Courseware and the assessment are on the LMS: <https://lms-tms.tertiaryinfotech.com/>

Digital Attendance



Attendance is mandatory for SSG funding.

Take AM, PM and Assessment digital attendance for every session.

Use the SSG / TMS QR code provided by your trainer.

**Minimum
75%
attendance
required**

Thank You!

Questions? We're here to help — during and after class.

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